RACE
Regular Analysis Through Continuous Exercise

Vector : RACE-01

Single Correct Question

1. The position vectors of two points A and B are $\hat{i}+\hat{j}-\hat{k}$ and $2\hat{i}-\hat{j}+\hat{k}$ respectively. Then $|\overrightarrow{AB}|=$
(A) 2 (B) 3 (C) 4 (D) 5

MVT164

2. The position vectors of P and Q are $5\hat{i}+4\hat{j}+a\hat{k}$ and $-\hat{i}+2\hat{j}-2\hat{k}$ respectively. If the distance between them is 7, then the value of a will be
(A) –5, 1 (B) 5, 1 (C) 0, 5 (D) 1, 0

MVT165

3. If $\vec{a}$ and $\vec{b}$ are two vectors of magnitude 1 inclined at 120°, then the angle between $\vec{b}$ and $\vec{b}-\vec{a}$ is-

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) $\frac{5\pi}{6}$

MVT166

4. If $ABCD$ is a parallelogram, $\overrightarrow{AB}=2\hat{i}+4\hat{j}-5\hat{k}$ and $\overrightarrow{AD}=\hat{i}+2\hat{j}+3\hat{k}$, then the unit vector in the direction of BD is

(A) $\frac{1}{\sqrt{69}}(\hat{i}+2\hat{j}-8\hat{k})$ (B) $\frac{1}{69}(\hat{i}+2\hat{j}-8\hat{k})$ (C) $\frac{1}{\sqrt{69}}(-\hat{i}-2\hat{j}+8\hat{k})$ (D) $\frac{1}{69}(-\hat{i}-2\hat{j}+8\hat{k})$

Rankers' Academy JEE

MVT167

Subjective Type Question

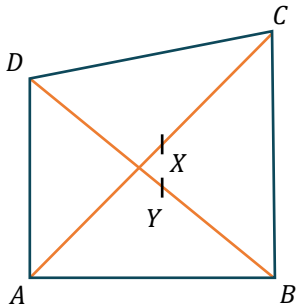
5. Let point A, B, C, D are vertices of parallelogram $ABCD$. Then prove that $2\overrightarrow{AB}+3\overrightarrow{BC}+4\overrightarrow{CD}+5\overrightarrow{DA}=2\overrightarrow{CA}$.

MVT168

6. Let O be the centre of a regular pentagon $ABCDE$ and $\overrightarrow{OA}=\vec{a}$. Then prove that $\overrightarrow{AB}+2\overrightarrow{BC}+3\overrightarrow{CD}+4\overrightarrow{DE}+5\overrightarrow{EA}$ equals $5\vec{a}$.

MVT169

7. In figure, X and Y are the mid-points of AC and BD. Write down an expression for $\overrightarrow{AY}$ in terms of $\overrightarrow{AD}$ and $\overrightarrow{AB}$. Write down an expression for $\overrightarrow{YX}$ in terms of $\overrightarrow{YA}$ & $\overrightarrow{YC}$. Show that $\overrightarrow{AB}+\overrightarrow{AD}+\overrightarrow{CB}+\overrightarrow{CD}=4\bullet\overrightarrow{XY}$.



MVT170

For More Join: @IITJEE_Advanced

8. If D, E and F are the mid-points of the sides BC, CA and AB, respectively of ΔABC and O is any point, then prove the following.

(a) $\overrightarrow{OA}+\overrightarrow{OB}+\overrightarrow{OC}=\overrightarrow{OD}+\overrightarrow{OE}+\overrightarrow{OF}$ (b) $\overrightarrow{AD}+\overrightarrow{BE}+\overrightarrow{CF}=0$

(c) $\overrightarrow{OE}+\overrightarrow{OF}+\overrightarrow{DO}=\overrightarrow{OA}$ (d) $\overrightarrow{AD}+\frac{2}{3}\overrightarrow{BE}+\frac{1}{3}\overrightarrow{CF}=\frac{1}{2}\overrightarrow{AC}$

MVT171

9. If M and N are the mid points of the sides BC and CD respectively of a parallelogram $ABCD$, then $\overline{AM}+\overline{AN}=\lambda\overline{AC}$; where λ is a scalar. Find λ.

MVT172

10. If $\vec{a}=2\hat{i}-2\sin\theta\hat{j}+3\hat{k}$ and $\vec{b}=4\cos\alpha\hat{i}-\hat{j}+2\sqrt{3}\cos\theta\hat{k}$ are equal vectors then find general values of 'θ' & 'α'.

MVT173

11. Find all values of λ such that $(x,y,z)\neq(0,0,0)$ and $\left(\vec{i}+\vec{j}+3\vec{k}\right)x+\left(3\vec{i}-3\vec{j}+\vec{k}\right)y+\left(-4\vec{i}+5\vec{j}\right)z=\lambda\left(\vec{i}x+\vec{j}y+\vec{k}z\right)$, where $\vec{i},\vec{j},\vec{k}$ are unit vectors along the coordinate axes.

MVT174

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	9	10	11
Answer	B	A	C	C	$\frac{3}{2}$	$\theta=2n\pi+\frac{\pi}{6}$, $\alpha=2n\pi\pm\frac{\pi}{3}$	(0, –1)

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Vector : RACE-02

Multiple Correct Question

1. A line passes through a point A with position vector $3\hat{i}+\hat{j}-\hat{k}$ and is parallel to the vector $2\hat{i}-\hat{j}+2\hat{k}$. If P is a point on this line such that $AP = 15$ units, then the position vector of the point P is/are

(A) $13\hat{i}+4\hat{j}-9\hat{k}$ (B) $13\hat{i}-4\hat{j}+9\hat{k}$ (C) $7\hat{i}-6\hat{j}+11\hat{k}$ (D) $-7\hat{i}+6\hat{j}-11\hat{k}$

MVT175

Subjective Type Question

2. Find the position vectors of the points which divide the join of the points $2\vec{a}-3\vec{b}$ and $3\vec{a}-2\vec{b}$ internally and externally in the ratio 2 : 3.

MVT176

3. P, Q have position vectors $\vec{a}$ and $\vec{b}$ relative to the origin 'O' and X, Y divide $\overrightarrow{PQ}$ internally and externally respectively in the ratio 2 : 1. Find vector $\overrightarrow{XY}$ in terms of $\vec{a},\vec{b}$ -

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MVT177

4. Let $\vec{p}$ is the p.v. of the orthocentre & $\vec{g}$ is the p.v. of the centroid of the triangle ABC where circumcentre is at origin. If $\vec{p} = K\vec{g}$, then find value of K.

MVT178

5. Let S, G and H be the circumcenter, centroid and orthocenter respectively of triangle ABC. Then prove the following :

(i) $\overrightarrow{PA}+\overrightarrow{PB}+\overrightarrow{PC}=3\overrightarrow{PG}$, where P is any point in the plane of triangle ABC.

(ii) $\overrightarrow{SA}+\overrightarrow{SB}+\overrightarrow{SC}=\overrightarrow{SH}$.

(iii) $\overrightarrow{HA}+\overrightarrow{HB}+\overrightarrow{HC}=2\overrightarrow{HS}$.

MVT179

6. If the vectors $\vec{a}=3\hat{i}+\hat{j}-2\hat{k}$, $\vec{b}=-\hat{i}+3\hat{j}+4\hat{k}$ & $\vec{c}=4\hat{i}-2\hat{j}-6\hat{k}$ constitute the sides $\overrightarrow{CB},\overrightarrow{CA}$ & $\overrightarrow{AB}$ of a ΔABC, then find

(a) The length of the median bisecting the vector $\vec{c}$.

(b) Length of internal angle bisector of angle A.

MVT180

7. The position vectors of the points A, B, C where ABC is a triangle are $\vec{a},\vec{b}$ and $\vec{c}$ respectively. E is a point in AB such that $AE : EB = 2 : 3$. Find the position vector of the point P where CE meets the median AD in terms of the vectors $\vec{a},\vec{b}$ and $\vec{c}$.

MVT181

For More Join: @IITJEE_Advanced

8. Find the vector equation of the line which is parallel to the vector $3\hat{i}-2\hat{j}+6\hat{k}$ and which passes through the point (1, –2, 3).

MVT182

9. Find the unit vectors along the angle bisector between the vectors $\vec{a}=\hat{i}+2\hat{j}-2\hat{k}$ and $\vec{b}=-3\hat{i}+6\hat{j}+2\hat{k}$.

MVT183

10. Find point of intersection of lines $\vec{r}=\left(\hat{i}+2\hat{j}+3\hat{k}\right)+\lambda\left(2\hat{i}+4\hat{j}+5\hat{k}\right)$ and

$\vec{r}=3\hat{i}+6\hat{k}+8\hat{k}+\mu\left(6\hat{i}-7\hat{j}+8\hat{k}\right)$

MVT184

11. Find point of intersection of line $\vec{r}=2\hat{i}-\hat{j}-3\hat{k}+\lambda\left(-\hat{i}+2\hat{j}+2\hat{k}\right)$ and plane

$\vec{r}=\left(5\hat{i}-7\hat{j}-9\hat{k}\right)+\mu\left(\hat{i}-2\hat{j}\right)+\nu\left(3\hat{k}-\hat{j}\right)$

MVT185

12. Let $\vec{r}=\vec{a}+\lambda\vec{l}$ and $\vec{r}=\vec{b}+\mu\vec{m}$ be two lines in space where $\vec{a}=5\hat{i}+\hat{j}+2\hat{k}$, $\vec{b}=-\hat{i}+7\hat{j}+8\hat{k}$, $\vec{l}=-4\hat{i}+\hat{j}-\hat{k}$ and $\vec{m}=2\hat{i}-5\hat{j}-7\hat{k}$ then find the p.v. of a point which lies on both of these lines.

MVT186

Rankers Academy JEE

ANSWER KEY

1.	B,D	**2.**	$\left(-5\vec{b}\right)$	**3.**	$\frac{4\vec{b}-4\vec{a}}{3}$	**4.**	3
6.	(a) $\sqrt{6}$ (b) $\frac{3}{4}\sqrt{10}$	**7.**	$\frac{3\vec{a}}{7}+\frac{2\vec{b}}{7}+\frac{2\vec{c}}{7}$	**8.**	$\vec{r}=\left(\hat{i}-2\hat{j}+3\hat{k}\right)+\lambda(3\hat{i}-2\hat{j}+6\hat{k})$		
9.	$\frac{-\hat{i}+16\hat{j}-4\hat{k}}{\sqrt{273}}$	**10.**	(3, 6, 8)	**11.**	(5, –7, –9)		
12.	(1, 2, 1)						

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Vector : RACE-03

Subjective Type Question

1. Find the angle between two vectors $\vec{a}$ & $\vec{b}$ with magnitude 2 and 1 respectively and such that $\vec{a}.\vec{b} = \sqrt{3}$.

MVT187

2. If $\vec{a}=2\hat{i}+2\hat{j}+3\hat{k}$, $\vec{b}=-\hat{i}+2\hat{j}+\hat{k}$ and $\vec{c}=3\hat{i}+\hat{j}$ are such that $\vec{a}+\lambda\vec{b}$ perpendicular to $\vec{c}$ then the value of λ is

MVT188

3. The scalar product of the vector $\hat{i}+\hat{j}+\hat{k}$ with a unit vector along the sum of the vectors $2\hat{i}+4\hat{j}-5\hat{k}$ and $\lambda\hat{i}+2\hat{j}+3\hat{k}$ is equal to 1, find λ.

MVT189

4. If the vectors $3\bar{p}+\bar{q}$; $5\bar{p}-3\bar{q}$ and $2\bar{p}+\bar{q}$; $4\bar{p}-2\bar{q}$ are pairs of mutually perpendicular vectors then find $\sin(\bar{p}\wedge\bar{q})$.

MVT190

5. Find the acute angle between the medians drawn from the acute angles of an isosceles right angled triangle.

Rankers Academy JEE

MVT191

6. Find the set of values of c for which the angle between the vectors $cx\hat{i}-6\hat{j}+3\hat{k}$ & $x\hat{i}-2\hat{j}+2cx\hat{k}$ is acute for every $x \in R$.

MVT192

7. Let $\vec{a}, \vec{b}, \vec{c}$ be vectors of length 3, 4, 5 respectively. Let $\vec{a}$ be perpendicular to $\vec{b}+\vec{c}$, $\vec{b}$ to $\vec{c}+\vec{a}$ & $\vec{c}$ to $\vec{a}+\vec{b}$. Then find $\left|\vec{a}+\vec{b}+\vec{c}\right|$.

MVT193

8. Let $\vec{u}=\hat{i}+\hat{j}$, $\vec{v}=\hat{i}-\hat{j}$ and $\vec{w}=\hat{i}+2\hat{j}+3\hat{k}$ If $\hat{n}$ is a unit vector such that $\vec{u}\cdot\hat{n}=0$ and $\vec{v}\cdot\hat{n}=0$, then find $|\vec{w}\cdot\hat{n}|$.

MVT194

9. Find the projection of the vector $\vec{a}=4\hat{i}-2\hat{j}+\hat{k}$ on the vector $\vec{b}=3\hat{i}+6\hat{j}+2\hat{k}$. Also find vector component of along $\vec{a}$ and $\vec{b}$ perpendicular to $\vec{b}$.

MVT195

10. Prove that in a regular tetrahedron any pair of opposite edges are perpendicular.

MVT196

11. If $2\sin x \sin y + 3\cos y + 6\cos x \sin y = 7$, then find the value of $\tan^2 x + 2\tan^2 y$.

MVT197

For More Join: @IITJEE_Advanced

12. Given a parallelogram $ABCD$. If $|\vec{AB}| = a$, $|\vec{AD}| = b$ & $|\vec{AC}| = c$, then find $\vec{DB} \cdot \vec{AB}$.

MVT198

13. In the isosceles triangle ABC $|\overrightarrow{AB}| = |\overrightarrow{BC}| = 8$, a point E divides AB internally in the ratio 1 : 3, then find the cosine of the angle between $\overrightarrow{CE}$ & $\overrightarrow{CA}$ (where $|\overrightarrow{CA}| = 12$)

MVT199

14. L_1 and L_2 are two lines whose vector equations are

$L_1 : \vec{r} = \lambda\left[\left(\cos\theta + \sqrt{3}\right)\hat{i} + \left(\sqrt{2}\sin\theta\right)\hat{j} + \left(\cos\theta - \sqrt{3}\right)\hat{k}\right]$

$L_2 : \vec{r} = \mu\left(a\hat{i} + b\hat{j} + c\hat{k}\right)$,

where λ and μ are scalars and α is the angle between L_1 and L_2. If the angle 'α' is independent of θ then find the value of 'α'

MVT200

Rankers Academy JEE

ANSWER KEY

1.	$\frac{\pi}{6}$	**2.**	8	**3.**	1	**4.**	$\frac{\sqrt{55}}{8}$
5.	$\cos^{-1}\frac{4}{5}$	**6.**	$c \in \left[0, \frac{4}{3}\right)$	**7.**	$5\sqrt{2}$	**8.**	3
9.	$\frac{190\hat{i} - 110\hat{j} + 45\hat{k}}{49}$			**11.**	9	**12.**	$\frac{3a^2 - c^2 + b^2}{2}$
13.	$\frac{3\sqrt{7}}{8}$	**14.**	30°				

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Vector : RACE-04

Single Correct Question

1. If $\vec{a}=(1,-1,1)$ and $\vec{c}=(-1,-1,0)$, then the vector $\vec{b}$ satisfying $\vec{a}\times\vec{b}=\vec{c}$ and $\vec{a}\,.\,\vec{b}=1$ is

(A) (1, 0, 0) (B) (0, 0, 1) (C) (0, –1, 0) (D) None of these

MVT201

2. For two particular vectors $\vec{A}$ and $\vec{B}$ it is known that $\vec{A}\times\vec{B} = \vec{B}\times\vec{A}$. What must be true about the two vectors?

(A) At least one of the two vectors must be the zero vector.

(B) $\vec{A}\times\vec{B} = \vec{B}\times\vec{A}$ is true for any two vectors.

(C) One of the two vectors is a scalar multiple of the other vector.

(D) The two vectors must be perpendicular to each other.

MVT202

3. Find the number of vectors of unit length perpendicular to vectors $\vec{a}$ = (1, 1, 0) & $\vec{b}$ (0, 1, 1).

(A) 2 (B) 3 (C) 4 (D) ∞

MVT203

Rankers Academy JEE

Multiple Correct Question

4. Let $\vec{a},\vec{b},\vec{c}$ be three vectors such that $\vec{a}\cdot\vec{a}=\vec{b}\cdot\vec{b}=\vec{c}\cdot\vec{c}=3$ and $\left|\vec{a}+\vec{b}-\vec{c}\right|^2+\left|\vec{b}+\vec{c}-\vec{a}\right|^2+\left|\vec{c}+\vec{a}-\vec{b}\right|^2=36$, then

(A) $\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}=\dfrac{-9}{2}$ (B) $\vec{a},\vec{b},\vec{c}$ are coplanar vectors

(C) $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}\neq\vec{0}$ (D) $\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}=\dfrac{-27}{2}$

MVT204

Subjective Type Question

5. If $\vec{a}$ and $\vec{b}$ are vectors such that $\left|\vec{a}+\vec{b}\right|=\sqrt{29}$ and $\vec{a}\times(2\hat{i}+3\hat{j}+4\hat{k}) = (2\hat{i}+3\hat{j}+4\hat{k})\times\vec{b}$, value of $\left|(\vec{a}+\vec{b}).(-7\hat{i}+2\hat{j}+3\hat{k})\right|$ is

MVT205

6. Find the area of the parallelogram whose diagonals are represented by the vectors $2\hat{i}+3\hat{j}-6\hat{k}$ and $3\hat{i}-4\hat{j}-\hat{k}$.

MVT206

7. If $\vec{a}\times\vec{b} = \vec{c}\times\vec{d}$ and $\vec{a}\times\vec{c} = \vec{b}\times\vec{d}$, then show that $(\vec{a}-\vec{d})$ is parallel to $(\vec{b}-\vec{c})$ when $\vec{a}\neq\vec{d}$ and $\vec{b}\neq\vec{c}$.

MVT207

For More Join: @IITJEE_Advanced

8. For any two vectors, $\vec{u}$ & $\vec{v}$ prove that

(a) $(\vec{u}.\vec{v})^2+|\vec{u}\times\vec{v}|^2=|\vec{u}|^2|\vec{v}|^2$ (b) $(1+|\vec{u}|^2)(1+|\vec{v}|^2)=(1-\vec{u}.\vec{v})^2+|\vec{u}+\vec{v}+(\vec{u}\times\vec{v})|^2$

MVT208

9. If $\vec{A},\vec{B},\vec{C}$ are unit vectors. Suppose that $\vec{A}.\vec{B}=\vec{A}.\vec{C}=0$ and the angle between $\vec{B}$ and $\vec{C}$ is $\frac{\pi}{6}$. Prove that $\vec{A}=\pm2\left(\vec{B}\times\vec{C}\right)$.

MVT209

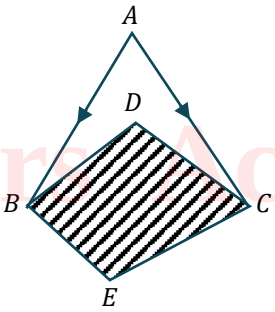
10. If A,B,C,D are any four points in space, prove that $\left|\overrightarrow{AB}\times\overrightarrow{CD}+\overrightarrow{BC}\times\overrightarrow{AD}+\overrightarrow{CA}\times\overrightarrow{BD}\right|=4$ (Area of the triangle ABC)

MVT210

11. A_1, A_2,....., A_n are the vertices of a regular plane polygon with n sides and O is its centre. Show that $\sum_{i=1}^{n-1}\left(\overrightarrow{OA_i}\times\overrightarrow{OA_{i+1}}\right)=(1-n)\left(\overrightarrow{OA_2}\times\overrightarrow{OA_1}\right)$

MVT211

12. Let $\overrightarrow{AB}=3\hat{i}-\hat{j}$, $\overrightarrow{AC}=2\hat{i}+3\hat{j}$ and $\overrightarrow{DE}=4\hat{i}-2\hat{j}$. Then, find the area of the shaded region in the figure



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MVT212

13. Position vectors of the four angular points of a tetrahedron $ABCD$ are $A(3, -2, 1)$; $B(3, 1, 5)$; $C(4, 0, 3)$ and $D(1, 0, 0)$. Find the acute angle between the plane faces ADC and ABC.

MVT213

ANSWER KEY

Question	1	2	3	4	5	6	12	13	
Answer	B	C	A	ABC	4	$\frac{7\sqrt{26}}{2}$	7	$\theta=\tan^{-1}\frac{5}{2}$	

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Vector : RACE-05

Single Correct Question

1. Let $\vec{a}=\hat{i}+\hat{j}+\hat{k}, \vec{b}=\hat{i}-\hat{j}+\hat{k}$ and $\vec{c}=\hat{i}-\hat{j}-\hat{k}$ be three vectors. A vector $\vec{v}$ in the plane of $\vec{a}$ and $\vec{b}$, whose projection on $\vec{c}$ is $\frac{1}{\sqrt{3}}$, is given by

(A) $\hat{i}-3\hat{j}+3\hat{k}$ (B) $-3\hat{i}-3\hat{j}-\hat{k}$ (C) $3\hat{i}-\hat{j}+3\hat{k}$ (D) $\hat{i}+3\hat{j}-3\hat{k}$

MVT214

Subjective Type Question

2. If $\vec{u}, \vec{v}, \vec{w}$ are non-coplanar vectors and p, q are integers, then find the number of ordered pairs (p, q) for which the equality $[3\vec{u}\ \ p\vec{v}\ \ p\vec{w}] - [p\vec{v}\ \ \vec{w}\ \ q\vec{u}] - [2\vec{w}\ \ q\vec{v}\ \ q\vec{u}] = 0$ holds.

MVT215

3. If $\vec{r}$ be a vector perpendicular to $\vec{a}+\vec{b}+\vec{c}$, where $\left[\vec{a}\ \vec{b}\ \vec{c}\right] \neq 0$ and $\vec{r} = \ell(\vec{b} \times \vec{c}) + m(\vec{c} \times \vec{a}) + n(\vec{a} \times \vec{b})$, then find $l + m + n$.

MVT216

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4. If $\vec{a}, \vec{b}$ and $\vec{c}$ are non zero, non coplanar vectors determine whether the vectors $\vec{r}_1=2\vec{a}-3\vec{b}+\vec{c}$, $\vec{r}_2=3\vec{a}-5\vec{b}+2\vec{c}$ and $\vec{r}_3=4\vec{a}-5\vec{b}+\vec{c}$ are linearly independent or dependent.

MVT217

5. Find a vector $\vec{r}$ in the plane of $\vec{p}=-\hat{i}+\hat{j}$ and $\vec{q}=-\hat{j}+\hat{k}$ such that $\vec{r}$ is perpendicular to $\vec{p}$ and $\vec{r}.\vec{q}=-2$.

MVT218

6. Let $\vec{a}=-\hat{i}-\hat{k}$, $\vec{b}=-\hat{i}+\hat{j}$ and $\vec{c}=\hat{i}+2\hat{j}+3\hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r}\times\vec{b}=\vec{c}\times\vec{b}$ and $\vec{r}\ .\ \vec{a}=0$, then the value of $\vec{r}\ .\ \vec{b}$ is

MVT219

7. Suppose $\vec{v}_1,\vec{v}_2,\vec{v}_3$ is an orthogonal set of vectors in $\mathbb{R}^3$ with $\vec{v}_1.\vec{v}_1$ = 29, $\vec{v}_2.\vec{v}_2$ = 85.25, $\vec{v}_3.\vec{v}_3$ = 9, $\vec{w}$ is another vector in same space such that $\vec{w}.\vec{v}_1$ = –29, $\vec{w}.\vec{v}_2$ = 341, $\vec{w}.\vec{v}_3$ = –18, then find $\vec{w}$ in terms of $\vec{v}_1,\vec{v}_2,\vec{v}_3$.

MVT220

8. Find volume of parallelopiped formed by vectors $2\hat{i}-3\hat{j}+4\hat{k}, 3\hat{j}+2\hat{j}-4k, 4\hat{i}+3\hat{j}-2\hat{k}$;

MVT221

9. Find the volume of tetrahedron $ABCD$ with edge lengths AB = 2, AC 3, AD = 4, BC = $\sqrt{13}$, BD = $2\sqrt{5}$ and CD = 5.

MVT222

For More Join: @IITJEE_Advanced

10. Let M be the midpoint of AB in regular tetrahedron $ABCD$. What is $\cos(\angle CMD)$?

MVT223

11. Find the shortest distance between the lines :

$\vec{r}_1 = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k})$ & $\vec{r}_2 = (2\hat{i} + 4\hat{j} + 5\hat{k}) + \mu(3\hat{i} + 4\hat{j} + 5\hat{k})$.-

MVT224

12. Find the value of a and point of intersection of lines

$\vec{r} = \left(\hat{i} + 2\hat{j} + \hat{k}\right) + \lambda\left(\hat{i} - \hat{j} + \hat{k}\right)$ and $\vec{r} = 2\hat{i} - \hat{j} - \hat{k} + \mu\left(2\hat{i} + \hat{j} + a\hat{k}\right)$, given that they intersect.

MVT225

13. A line passes through the point $A\left(\hat{i} + 2\hat{j} + 3\hat{k}\right)$ and is parallel to the vector $\vec{V}\left(\hat{i} + \hat{j} + \hat{k}\right)$. Then find its shortest distance from the origin.

MVT226

Rankers Academy JEE

ANSWER KEY

1.	C	**2.**	1	**3.**	0	**4.**	linearly dependent	**5.**	$\frac{2}{3}\left(\hat{i} + \hat{j} - 2\hat{k}\right)$
6.	9	**7.**	$-\vec{v}_1 + 4\vec{v}_2 - 2\vec{v}_3$			**8.**	50	**9.**	4
10.	$\frac{1}{3}$	**11.**	$\frac{1}{\sqrt{6}}$			**12.**	$a = \frac{13}{2}$ and $\left(\frac{10}{3}, \frac{-1}{3}, \frac{10}{3}\right)$	**13.**	$\sqrt{2}$

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Vector : RACE-06

Subjective Type Question

1. Find the volume of the parallelopiped whose coterminous edges are represented by $\vec{a}=2\hat{i}-3\hat{j}+4\hat{k}, \vec{b}=\hat{i}+2\hat{j}-\hat{k}$ and $\vec{c}=3\hat{i}-\hat{j}+2\hat{k}$.

MVT227

2. The volume of the tetrahedron formed by the coterminus edges $\vec{a}, \vec{b}, \vec{c}$ is 3. Then find the volume of the parallelepiped formed by the coterminus edges $\vec{a}+\vec{b}, \vec{b}+\vec{c}, \vec{c}+\vec{a}$.

MVT228

3. If $\vec{a} = 2\hat{i} - 4\hat{j} + 7\hat{k}$, $\vec{b} = 3\hat{i} + 5\hat{j} - 9\hat{k}$ and $\vec{c} = \hat{i} + \hat{j} + \hat{k}$, then find $[\vec{a}\ \vec{b}\ \vec{c}]$ and also $\vec{a} \times (\vec{b} \times \vec{c})$.

MVT229

4. $\vec{a},\vec{b}$ and $\vec{c}$ be three vectors having magnitudes 1, 1 and 2 respectively. If $\vec{a}\times(\vec{a}\times\vec{c})+\vec{b}=0$, then find the acute angle between $\vec{a}$ & $\vec{c}$.

MVT230

Rankers Academy JEE

5. Let $\vec{\alpha}=2\hat{i}+3\hat{j}-\hat{k}$ and $\vec{\beta}=\hat{i}+\hat{j}$. If $\vec{\gamma}$ is a unit vector, then find the maximum value of $\left[\vec{\alpha}\times\vec{\beta}\ \ \vec{\beta}\times\vec{\gamma}\ \ \vec{\gamma}\times\vec{\alpha}\right]$.

MVT231

6. If $\vec{a}=\frac{1}{\sqrt{10}}(3\hat{i}+\hat{k})$ and $\vec{b}=\frac{1}{7}(2\hat{i}+3\hat{j}-6\hat{k})$, then find the value of $(2\vec{a}-\vec{b}).\left[(\vec{a}\times\vec{b})\times(\vec{a}+2\vec{b})\right]$.

MVT232

7. Given the following information about the non-zero vectors $\vec{A},\vec{B}$ and $\vec{C}$:

(i) $(\vec{A}\times\vec{B})\times\vec{A}=\vec{0}$ (ii) $\vec{B}.\vec{B}=4$ (iii) $\vec{A}.\vec{B}=-6$ (iv) $\vec{B}.\vec{C}=6$

Then find the value of $\left|\vec{A}\times\vec{B}\right|+\left|\vec{A}.\left(\vec{B}\times\vec{C}\right)\right|+\left|\vec{A}.\vec{C}\right|$.

MVT233

8. Let $\vec{a},\vec{b}$ and $\vec{c}$ are non-zero unit vectors such that they are not-orthogonal pairwise.

If $\vec{V}_1=\vec{a}\times(\vec{b}\times\vec{c})$ and $\vec{V}_2=(\vec{a}\times\vec{b})\times\vec{c}$ are such that $\vec{V}_1=\vec{V}_2$, then find

$\left|(\vec{b}\times\vec{c}).(\vec{c}\times\vec{a})\right|+\left|(\vec{c}\times\vec{a}).(\vec{a}\times\vec{b})\right|+\left|(\vec{c}\times\vec{a})\times(\vec{b}\times\vec{c})\right|$.

MVT234

9. If $\vec{a}.\vec{b}=0, |\vec{a}|=1$ and $|\vec{b}|=\frac{1}{2}$, then find $\left|\vec{a}\times\left(\vec{a}\times\left(\vec{a}\times(\vec{a}\times\vec{b})\right)\right)\right|$.

MVT235

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10. If $\vec{A}=2\hat{i}+3\hat{k},\vec{B}=-2\hat{i}+\hat{j}+\hat{k},\vec{C}=10\hat{j}-\hat{k}$ determine x,y,z so that $\left(\vec{A}\times\vec{B}\right)\times\vec{C}=x\vec{A}+y\vec{B}+z\vec{C}$

MVT236

11. If $\vec{a},\vec{b},\vec{c}$ are unit vectors, with $\vec{a}\wedge\vec{b}=\frac{\pi}{3}=\vec{b}\wedge\vec{c}=\vec{c}\wedge\vec{a}$ then find $\left[\vec{a}\times\vec{b}\quad\vec{b}\times\vec{c}\quad\vec{c}\times\vec{a}\right]$.

MVT237

12. Vector $\vec{x},\vec{y},\vec{z}$ each of magnitude $\sqrt{2}$, make angles of 60° with each other. If $\vec{x}\times\left(\vec{y}\times\vec{z}\right)=\vec{a}$, $\vec{y}\times\left(\vec{z}\times\vec{x}\right)=\vec{b}$ and $\vec{x}\times\vec{y}=\vec{c}$ then find $\vec{x},\vec{y},\vec{z}$ in terms of $\vec{a},\vec{b},\vec{c}$.

MVT238

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ANSWER KEY

1.	7	**2.**	36	**3.**	$62,\ 92\hat{i}+102\hat{j}+32\hat{k}$	**4.**	$\frac{\pi}{6}$	**5.**	3	**6.**	–5
7.	9	**8.**	0	**9.**	$\frac{1}{2}$	**10.**	$x=-9, y=-3, z=0$				
11.	$\frac{1}{2}$	**12.**	$x=\frac{(\vec{a}+\vec{b})\times\vec{c}+\vec{a}-\vec{b}}{-2};\ y=\frac{(\vec{a}+\vec{b})\times\vec{c}+\vec{a}+\vec{b}}{-2};\ \vec{z}=\frac{(\vec{a}+\vec{b})\times\vec{c}+\vec{a}-\vec{b}}{-2}$								

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RACE
Regular Analysis Through Continuous Exercise

Vector : RACE-07

Single Correct Question

1. Let $\vec{a}=\hat{i}-2\hat{j}+\hat{k}$ and $\vec{b}=\hat{i}-\hat{j}+\hat{k}$ be two vectors. If $\vec{c}$ is a vector such that $\vec{b}\times\vec{c}=\vec{b}\times\vec{a}$ and $\vec{c}.\vec{a}=0$, then $\vec{c}.\vec{b}$ is equal to

(A) $\frac{1}{2}$ (B) –1 (C) $-\frac{1}{2}$ (D) $-\frac{3}{2}$

MVT239

2. Let $\vec{a}=\hat{i}-\hat{j},\vec{b}=\hat{i}+\hat{j}+\hat{k}$ and $\vec{c}$ be a vector such that $\vec{a}\times\vec{c}+\vec{b}=\vec{0}$ and $\vec{a}.\vec{c}=4$, then $|\vec{c}|^2$ is equal to

(A) $\frac{19}{2}$ (B) 8 (C) $\frac{17}{2}$ (D) 9

MVT240

Subjective Type Question

3. If $\vec{p}=\frac{\vec{b}\times\vec{c}}{[\vec{b}\ \vec{c}\ \vec{a}]},\vec{q}=\frac{\vec{c}\times\vec{a}}{[\vec{c}\ \vec{a}\ \vec{b}]},\vec{r}=\frac{\vec{a}\times\vec{b}}{[\vec{a}\ \vec{b}\ \vec{c}]}$, then find the value of $(\vec{a}+\vec{b}).\vec{p}+(\vec{b}+\vec{c}).\vec{q}+(\vec{c}+\vec{a})\cdot\vec{r}$.

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MVT241

4. If $\vec{a},\vec{b},\vec{c}$ are three non-coplanar vectors and $\vec{a}'=\frac{\vec{b}\times\vec{c}}{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]},\vec{b}'=\frac{\vec{c}\times\vec{a}}{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]}$ and $\vec{c}'=\frac{\vec{a}\times\vec{b}}{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]}$; then prove that

(a) $\vec{a}',\vec{b}',\vec{c}'$ are also not coplanar. (b) $\left(\vec{a}'\times\vec{b}'\right)+\left(\vec{b}'\times\vec{c}'\right)+\left(\vec{c}'\times\vec{a}'\right)=\frac{\vec{a}+\vec{b}+\vec{c}}{\left[\vec{a}\ \vec{b}\ \vec{c}\right]}$.

MVT242

5. For any vector $\vec{r}$, prove that $\vec{r}=(\vec{r}.\vec{a})\vec{a}'+\left(\vec{r}.\vec{b}\right)\vec{b}'+(\vec{r}.\vec{c})\vec{c}'$ and $\vec{r}=(\vec{r}.\vec{a}')\vec{a}+\left(\vec{r}.\vec{b}'\right)\vec{b}+(\vec{r}.\vec{c}')\vec{c}$.

MVT243

6. If $\hat{a},\hat{b},\hat{c}$ are three non-coplanar unit vectors and α,β,γ are the angles between $\hat{a}$ and $\hat{b}$, $\hat{b}$ and $\hat{c}$, $\hat{c}$ and $\hat{a}$ respectively and $\vec{x},\vec{y},\vec{z}$ are unit vectors along the bisectors of the angle α,β,γ respectively, then prove that $\left[\vec{x}\times\vec{y}\quad\vec{y}\times\vec{z}\quad\vec{z}\times\vec{x}\right]=\frac{1}{16}\left[a\quad b\quad c\right]^2\sec^2\frac{\alpha}{2}\sec^2\frac{\beta}{2}\sec^2\frac{\gamma}{2}$.-

MVT244

7. If $\vec{d}=\lambda\left(\vec{a}\times\vec{b}\right)+\mu\left(\vec{b}\times\vec{c}\right)+\nu\left(\vec{c}\times\vec{a}\right),\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]=\frac{1}{8}$ and $\vec{d}.\left(\vec{a}+\vec{b}+\vec{c}\right)=8$, then find $\lambda+\mu+\nu$.

MVT245

8. If $\vec{a}=\hat{i}+\hat{j}+\hat{k}$ & $\vec{b}=\hat{i}-2\hat{j}+\hat{k}$, then find the vector $\vec{c}$ such that $\vec{a}.\vec{c}=0$ & $\vec{a}\times\vec{c}=\vec{b}$.

MVT246

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9. Let $\vec{a}=\hat{j}-\hat{k}$ and $\vec{c}=\hat{i}-\hat{j}-\hat{k}$. Then find the vector $\vec{b}$ satisfying $\vec{a}\times\vec{b}+\vec{c}=\vec{0}$ and $\vec{a}\,.\,\vec{b}=3$.

MVT247

10. The vectors $\vec{a}$ and $\vec{b}$ are not perpendicular and $\vec{c}$ and $\vec{d}$ are two vectors satisfying $\vec{b}\times\vec{c}=\vec{b}\times\vec{d}$ and $\vec{a}.\vec{d}=0$. Then find the vector $\vec{d}$ in terms of $\vec{a},\vec{b},\vec{c}$.

MVT248

11. Let $ABCD$ be a parallelogram such that $\overrightarrow{AB}=\vec{q}$, $\overrightarrow{AD}=\vec{p}$ and $\angle BAD$ be an acute angle. If $\vec{r}$ is the vector that coincides with the altitude directed from the vertex B to the side AD, then find $\vec{r}$ in terms of $\vec{p}$ and $\vec{q}$.

MVT249

12. If $\vec{a},\vec{b}$ be two perpendicular unit vectors such that $\vec{x}=\vec{b}-(\vec{a}\times\vec{x})$, then find $|\vec{x}|$ -

MVT250

13. Let $\vec{u}=\hat{i}+\hat{j},\vec{v}=\vec{i}-\vec{j}$ and $\vec{w}=\vec{i}+2\vec{j}+3\vec{k}$. If $\vec{n}$ is a unit vector such that $\vec{u}.\vec{n}=0$ and $\vec{v}.\vec{n}=0$, then find $2|\vec{w}.\vec{n}|+|\vec{u}|^2+|\vec{v}|$.

MVT251

14. Given $|\vec{a}|=|\vec{b}|=1$ and $|\vec{a}+\vec{b}|=\sqrt{3}$. If '$c$' is a vector such that $\vec{c}-\vec{a}-2\vec{b}=3(\vec{a}\times\vec{b})$, then find $\vec{c}.\vec{b}$.

MVT252

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ANSWER KEY

1. C **2.** A **3.** 3 **7.** 64 **8.** $-\hat{i}+\hat{k}$ **9.** $\vec{b}=-\hat{i}+\hat{j}-2\hat{k}$

10. $\vec{d}=\vec{c}-\left(\frac{\vec{a}\cdot\vec{c}}{\vec{a}\cdot\vec{b}}\right)\vec{b}$ **11.** $\left(\frac{\vec{q}\cdot\vec{p}}{|\vec{p}|}\right)\vec{p}-\vec{q}$ **12.** $\frac{1}{\sqrt{2}}$ **13.** $8+\sqrt{2}$

14. $\frac{5}{2}$

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